

Shruti Singla (she/her)

3rd Year Computer Engineering Student at the University of British Columbia

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TECHNICAL SKILLS

Software: C/C++, Java, HTML/CSS, JavaScript, Python, Docker

Hardware: VHDL, Arduino, FPGA, Breadboarding, Oscilloscope

Tools & Platforms: Git, ROS 2, Model Sim, Quartus, IntelliJ IDEA, VS Code, Microsoft (outlook, word, excel)

EDUCATION

University of British Columbia

Expected Graduation: May 2027

Bachelor of Applied Science - Computer Engineering

Seneca College

January 2016 – January 2018

Diploma - Electronics Engineering Technician

TECHNICAL PROJECTS

Autonomous Robotics & Vision System | Robotics & AI Project

January 2026 – April 2026

- Developed an autonomous robotic car using ROS/ROS2, Python, LiDAR, and computer vision for real-time navigation.
- Implemented PID-based steering control and LiDAR-driven wall-following to maintain stable autonomous control.
- Trained a custom YOLO traffic light detection model using annotated [Roboflow](#) datasets without pretrained weights.
- Integrated OpenCV, and PyTorch for real-time traffic light classification and robotic decision-making.
- Developed ROS nodes and communication pipelines for sensor integration and motion control.
- Optimized navigation and detection performance under varying environmental conditions.

AI-Powered Collaborative Chat Application| UBC CPEN 320

January 2026 – April 2026

- Developed a full-stack real-time web application using HTML, CSS, JavaScript, Node.js, MongoDB, and WebSockets.
- Built backend functionality in Node.js to handle REST API requests, client communication, application logic, and database operations.
- Stored and managed persistent application data using MongoDB.
- Implemented real-time client-server messaging using WebSockets for live chat synchronization.
- Developed a prompt-engineered LLaMA 3.1 pipeline to extract action items, deadlines, and assignees from chat messages, returning structured JSON for downstream processing.
- Designed functionality to automatically highlight task-related messages and maintain a separate in-app task list.

Athlete Injury Prediction Model | Machine Learning Project

April 2026 – May 2026

- Developed a Python-based model to predict athlete injury risk using the Kaggle Injury Prediction Dataset.
- Implemented data preprocessing, feature analysis, and model evaluation using pandas, NumPy, scikit-learn, and PyTorch.
- Trained and compared Support Vector Machine, and Deep Neural Network models for binary injury classification.
- Evaluated model performance using accuracy, precision, recall, F1-score, and ROC analysis to balance prediction accuracy and interpretability.

WORK EXPERIENCE

First Service Residential, Vancouver, BC

August 2021 – Present

Concierge

- Collaborated with property management, security, and maintenance teams to support daily operations and deliver consistent resident service.
- Communicated clearly and professionally with residents, guests, and vendors to address inquiries, resolve issues, and maintain confidentiality.

A&W Restaurant, Toronto, ON

April 2016 – March 2021

Cashier and Supervisor

- Led and coordinated a team of 5–10 employees by delegating tasks, promoting teamwork, and maintaining service standards during peak hours.
- Trained, supported, and evaluated new staff to strengthen team performance, improve workflow efficiency, and uphold workplace standards.